SOUJANYA PONNAPALLI

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INTERESTS

Distributed systems · Storage systems · Operating systems · AI systems

I am a systems researcher. I build systems that are not just fast and scalable but also reliable, providing strong guarantees such as performance isolation and fault tolerance in practice.

WORK & EDUCATION

Postdoc · University of California, Berkeley — **UC Berkeley**
Supervisors: Prof. Natacha Crooks and Prof. Matei Zaharia — 2023-Ongoing

Research Intern · Microsoft Research, Redmond — **MSR**
Mentors: Jonathan Goldstein and Anirudh Badam — Summer'22, Summer'20

Research Intern · Microsoft Research, Cambridge — **MSR**
Mentors: Dushyanth Narayanan and Antony Rowstron — Summer'19

Research Intern · VMware Research, California — **VMware**
Mentors: Michael Wei and Dahlia Malkhi — Summer'18

PhD · University of Texas at Austin — **UT Austin**
Advisor: Prof. Vijay Chidambaram — 2017-23

Bachelor's · International Institute of Information Technology, Hyderabad — **IIIT-H**
Advisor: Prof. Suresh Purini — 2013-17

PUBLICATIONS

1. **Metronome: I/O-Efficient Logging for State Machine Replication** [VLDB'27]
   Under revision
2. **Delta Fair Sharing: Performance Isolation for Multi-Tenant Storage Systems** [VLDB'27]
   Under submission
3. **Time is Here for Just-in-Time Systems: Challenges and Opportunities** [MLForSys'26]
4. **Rollbaccine: Herd Immunity against Storage Rollback Attacks in TEEs** [SIGMOD'26]
5. **Supporting Our AI Overlords: Redesigning Data Systems to be Agent-First** [CIDR'26]
6. **SkyStore: Cost-Optimized Object Storage Across Regions and Clouds** [VLDB'25]
   [Best of VLDB'25 nomination]
7. **Real Life Is Uncertain. Consensus Should Be Too!** [HotOS'25]
8. **Lost in Translation: Network-Attached AI Accelerator Disaggregation** [HotNets'25]
9. **DINOMO: A Key-Value Store for Disaggregated Persistent Memory** [VLDB'22]
10. **RainBlock: Faster Transaction Processing in Public Blockchains** [ATC'21]
11. **WineFS: Hugepage-Aware File System for PM that Ages Gracefully** [SOSP'21]
12. **Software-Defined Data Protection at the Storage Layer** [VLDB'21]
13. **Finding Crash-Consistency Bugs with Bounded Black-Box Crash Testing** [OSDI'18]
14. **mLSM: Making Authenticated Storage Faster in Ethereum** [HotStorage'18]
15. **Holographic Storage for the Cloud: Advances and Challenges** [TOS'25]
16. **CrashMonkey and ACE: Systematically Testing File-System Crash Consistency** [TOS'19]

PREPRINTS

17. **Token Latency Fairness: Performance Isolation for Multi-Tenant LLM Serving** [arXiv'26]
18. **SKYE: Write-Optimized Key-Value Store for Persistent Memory** [arXiv'26]

PATENTS

Recoverable Processes [US Patent App. 17/981,296]

ONGOING PROJECTS

Page Cache Fairness: Performance Isolation for Multiple Tenants in Operating Systems
Powder: Fine-Grained Fault-Tolerance for Distributed Storage Systems
Fugue: Exploring Durability Trade-Offs in Replicated Storage

SERVICE

NSF Proposal Reviewer	2026
Program Committee	OSDI'26, NSDI'26, ATC'25, NSDI'25, EuroSys'25
External Program Committee	FAST'25, ATC'24, NSDI'19
Journal Reviewer, ACM TOS	2024

AWARDS

Summer Research Institute (SuRI) Fellowship, EPFL	2025
Berkeley RDI Frontier Research Funding	2024
The James C. Browne Graduate Fellowship, UT Austin	2017-18

REFERENCES

Natacha Crooks and Matei Zaharia (UC Berkeley), Vijay Chidambaram (UT Austin), Marcos K. Aguilera (NVIDIA), and Tianyin Xu (UIUC)